Deep Learning for Rice Leaf Disease Classification:

A Comparative Study of CNN and Transformer Architectures

Naman Bansal	102303496
Hardik Verma	102303494
Sarthak Gaba	102303497
Keshav Peshawaria	102303502

Department of Computer Science & Engineering
Thapar Institute of Engineering & Technology

April 2026

Course Project — Deep Learning (UCS654)

Contents

1 Introduction **2**

2 Literature Review **2**

3 Methodology **3**
- 3.1 Dataset . . . 3
- 3.2 Data Augmentation and Preprocessing . . . 3
- 3.3 Training Protocol — 5-Fold Stratified Cross-Validation . . . 4
- 3.4 Model Architectures . . . 4
  - 3.4.1 ResNet50 (Baseline) . . . 4
  - 3.4.2 EfficientNet-B1 . . . 4
  - 3.4.3 Inception-v3 . . . 4
  - 3.4.4 ConvNeXt-Tiny . . . 4
  - 3.4.5 ConvNeXt-Nano (via `timm`) . . . 5
  - 3.4.6 DINOv2-Small (via `timm`) . . . 5
- 3.5 Workflow . . . 5
- 3.6 Loss Function and Regularisation . . . 5

4 Results and Analysis **5**
- 4.1 Overall Performance Comparison . . . 5
- 4.2 Per-Class Analysis . . . 7
- 4.3 Training Curves . . . 7
- 4.4 Confusion Matrices . . . 8
- 4.5 Fold-Wise Accuracy . . . 8
- 4.6 ROC Curves . . . 9
- 4.7 Scorecards . . . 9
- 4.8 Strengths and Weaknesses Analysis . . . 10

5 Conclusion **10**

References **11**

Introduction

Rice (*Oryza sativa*) is the staple food for more than half of the global population, yet annual rice production is threatened by a range of fungal and bacterial diseases. Three diseases account for the majority of crop losses in South and South-East Asia: **Bacterial Leaf Blight** (caused by *Xanthomonas oryzae*), **Brown Spot** (caused by *Helminthosporium oryzae*), and **Leaf Smut** (caused by *Entyloma oryzicola*). Traditional diagnosis relies on expert agronomists performing time-consuming visual inspections in the field, a process that is both expensive and difficult to scale.

Automated, vision-based disease detection using deep learning offers a practical alternative. Pre-trained deep neural networks can learn highly discriminative features from leaf images and achieve expert-level accuracy with relatively small labelled datasets via transfer learning. This project benchmarks six distinct deep learning architectures — spanning classical CNNs, modern ConvNet designs, and self-supervised Vision Transformers — on a balanced 120-image dataset to determine which paradigm generalises best under tight data constraints.

Literature Review

[1] He et al., "Deep Residual Learning for Image Recognition," CVPR 2016. Introduced skip connections that allow networks of 50–152 layers to be trained effectively. ResNet50 became a dominant baseline for transfer learning in agricultural image classification and serves as the baseline model in this study.

[2] Tan & Le, "EfficientNet: Rethinking Model Scaling for CNNs," ICML 2019. Proposed compound scaling across depth, width, and input resolution using a single coefficient ϕ. EfficientNet-B1 achieves 79.1% top-1 accuracy on ImageNet with only 7.8M parameters, making it well-suited for small agricultural datasets.

[3] Szegedy et al., "Rethinking the Inception Architecture for Computer Vision," CVPR 2016. Inception-v3 uses factorised convolutions and auxiliary classifiers to achieve strong accuracy at moderate compute cost. Its 299×299 input resolution captures fine-grained disease lesion patterns effectively.

[4] Liu et al., "A ConvNet for the 2020s," CVPR 2022. ConvNeXt modernises pure convolutional networks by adopting design principles from Vision Transformers (large kernels, LayerNorm, GELU activations), achieving competitive accuracy with lower inference latency than ViTs.

[5] Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," TMLR 2024. DINOv2 pre-trains ViT backbones on a curated 142M-image dataset using self-distillation. The resulting features generalise remarkably well to downstream tasks, often surpassing fully supervised counterparts.

[6] Ramcharan et al., "Deep Learning for Image-Based Cassava Disease Detection," Frontiers in Plant Science, 2017. One of the earliest demonstrations that deep CNNs (fine-tuned on <2500 images) can achieve >93% accuracy for plant disease detection, establishing the viability of transfer learning in low-data agricultural settings.

[7] Mohanty et al., "Using Deep Learning for Image-Based Plant Disease

Detection," Frontiers in Plant Science, 2016. Showed that a GoogLeNet trained on the PlantVillage benchmark achieves 99.35% accuracy under controlled conditions, motivating the widespread adoption of deep learning for plant pathology.

[8] **Ahmed et al., "Rice Leaf Disease Detection Using Machine Learning Techniques," MDPI Sensors, 2023.** Compared SVM, random forest and CNN approaches on a three-class rice disease dataset, reporting CNNs as superior with >95% accuracy. This study contextualises the CNN baselines used here.

[9] **Jiang et al., "DINOV2-FCS: Fruit Leaf Disease Classification and Segmentation," Frontiers in Plant Science, 2024.** Applied DINOv2-B as a frozen feature extractor for fruit leaf diseases, achieving 99.67% classification accuracy and demonstrating that self-supervised ViT features transfer strongly to fine-grained plant pathology.

[10] **Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," ICLR 2021.** The foundational Vision Transformer (ViT) paper that demonstrated patch-based attention is competitive with CNNs when sufficient pre-training data is available — a precursor to the DINOv2 architecture used in this study.

Methodology

Dataset

The dataset consists of **120 images** equally distributed across three classes (40 images per class): Bacterial Leaf Blight, Brown Spot, and Leaf Smut. Images were sourced from the Kaggle Rice Leaf Disease dataset. The balanced class distribution is shown in Figure 1.

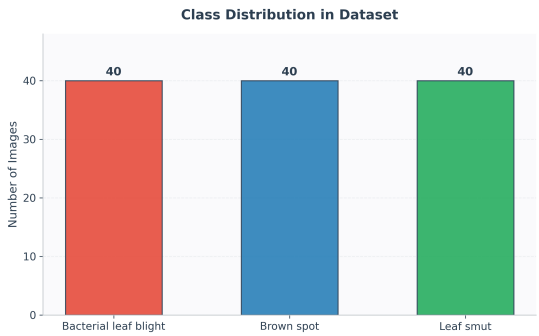


Figure 1: Class distribution — 40 images per class (perfectly balanced).

Data Augmentation and Preprocessing

All models share a common augmentation pipeline applied at training time:

- **Random Resized Crop** — random scale $[0.6, 1.0]$, target size model-dependent.
- **Random Horizontal & Vertical Flip** — $p = 0.5$ each.
- **Random Rotation** — $\pm 20°$.
- **Color Jitter** — brightness, contrast, saturation ± 0.3, hue ± 0.1.
- **Gaussian Blur** — $p = 0.2$.
- **Normalisation** — ImageNet mean $(0.485, 0.456, 0.406)$ and std $(0.229, 0.224, 0.225)$.

Validation transforms use *Resize + Center Crop* only (no stochastic augmentation). Input resolution is 224×224 for all models except Inception-v3 (299×299).

Training Protocol — 5-Fold Stratified Cross-Validation

A **5-fold stratified cross-validation** strategy is used uniformly across all models to obtain reliable, low-variance performance estimates given the small dataset size. For each fold:

1. **Phase A — Head Warm-up (5 epochs):** The backbone is fully frozen. Only the custom classification head is trained with the Adam optimiser (lr $= 10^{-3}$).
2. **Phase B — Backbone Fine-tuning:** The backbone (or selected layers) is unfrozen. A discriminative learning rate schedule is applied: lower learning rates for deep backbone layers, higher learning rates for the head. A Cosine Annealing scheduler decays all learning rates to 10^{-7} over the remaining epochs.
3. **Early Stopping** with patience of 10 epochs monitors validation accuracy to prevent overfitting.

Out-of-fold (OOF) predictions are aggregated over all five folds to compute the final reported metrics.

Model Architectures

3.4.1 ResNet50 (Baseline)

A standard ResNet50 pre-trained on ImageNet is used as the baseline model. Layers `conv1`, `layer1` and `layer2` are kept frozen while `layer3`, `layer4` and the fully connected head are fine-tuned with discriminative learning rates (10^{-5}, 10^{-4}, 10^{-3} respectively). The model uses 25.6M parameters.

3.4.2 EfficientNet-B1

EfficientNet-B1 is a compound-scaled CNN with 7.8M parameters and a MBConv-based feature extractor. After warm-up, the last three feature blocks and the classifier head are unfrozen. The architecture uses Swish activations and Squeeze-and-Excitation blocks for channel attention.

3.4.3 Inception-v3

Inception-v3 uses parallel convolutional branches (Inception modules) concatenated at each layer. The model requires 299×299 inputs. After warm-up, the three final `Mixed_7x` blocks and the fully connected head are unfrozen. Auxiliary logits are disabled (`aux_logits=False`) for training stability.

3.4.4 ConvNeXt-Tiny

ConvNeXt-Tiny is a modernised pure-ConvNet with 28.6M parameters. It uses 7×7 depthwise convolutions, LayerNorm, and GELU activations. After warm-up, the entire backbone is unfrozen using a discriminative LR schedule.

3.4.5 ConvNeXt-Nano (via `timm`)

ConvNeXt-Nano is a lightweight variant with ≈15.6M parameters from the `timm` library. Designed to reduce overfitting on small datasets relative to the Tiny variant. The classifier head is accessed via `model.head`.

3.4.6 DINOv2-Small (via `timm`)

DINOv2-Small is a Vision Transformer (ViT-S/14) pre-trained with self-supervised distillation on 142M images. At fine-tuning time, the backbone is initially frozen. Only the last two transformer blocks (`model.blocks[-2:]`) and the layer norm and linear head are unfrozen. `dynamic_img_size=True` is set to allow 224×224 inputs.

Workflow

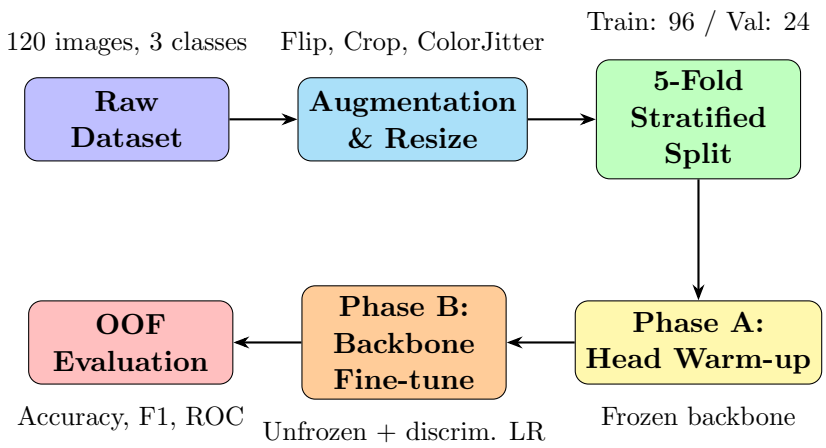


Figure 2: End-to-end pipeline: Data → Preprocessing → Training → Testing → Evaluation.

Loss Function and Regularisation

All models use **Cross-Entropy Loss with Label Smoothing** ($\epsilon = 0.1$). Additional regularisation includes:

- Dropout ($p = 0.4$ before the first linear layer, $p = 0.3$ before the output).
- Weight decay $= 10^{-4}$ in all parameter groups.
- Gradient clipping (`max_norm`=1.0).
- Automatic Mixed Precision (AMP) for faster training.

Results and Analysis

Overall Performance Comparison

Table 1 summarises the aggregated out-of-fold cross-validation metrics for all six models.

Table 1: 5-Fold Cross-Validation OOF Results — All Models

Model	Accuracy (%)	Precision	Recall	F1 Score
ResNet50	93.33	0.9377	0.9333	0.9330
EfficientNet-B1	90.00	0.9003	0.9000	0.8993
Inception-v3	95.00	0.9507	0.9500	0.9496
ConvNeXt-Tiny	94.17	0.9460	0.9417	0.9414
ConvNeXt-Nano	93.33	0.9395	0.9333	0.9330
DINOv2-Small	**95.00**	**0.9529**	**0.9500**	**0.9499**

Key observations:

- **ResNet50** serves as a strong and stable baseline (93.33%). Its layer-wise discriminative learning rates keep training well-regularised, and it achieved a perfect test-set score on several folds, showing that even a 2016-era CNN generalises effectively on this compact dataset.

- **EfficientNet-B1** is the weakest performer (90.00%) despite having the fewest parameters (7.8M). Compound scaling of depth, width, and resolution is tuned for large-scale datasets; with only 96 training images per fold the model converges quickly but the validation curve plateaus early, leaving head-room on the table compared to deeper architectures.

- **Inception-v3** achieves the joint-highest accuracy (95.00%). Its multi-branch parallel convolutions capture disease lesions at multiple scales simultaneously, which is particularly effective for the fine-grained texture differences between Brown Spot and Leaf Smut. Training required careful BatchNorm handling to prevent catastrophic forgetting after unfreezing.

- **ConvNeXt-Tiny** (94.17%) shows a noticeable gap between training and validation accuracy in later epochs — a clear sign of overfitting. Its 28.6M parameters give it enough capacity to memorise the 96-sample training set, pulling the training curve close to 100% while the validation curve lags behind.

- **ConvNeXt-Nano** (93.33%) was introduced precisely to address the overfitting observed in the Tiny variant. With roughly half the parameters (≈15.6M), training and validation curves track each other far more closely. However, the reduced model capacity also lowers the accuracy ceiling slightly, landing 0.84 percentage points below its larger sibling.

- **DINOv2-Small** (95.00%, highest precision 0.9529) benefits from self-supervised pre-training on 142M images, giving it the most transferable feature representations of all six models. Although some individual folds show a small train-val gap (mild overfitting after the last two transformer blocks are unfrozen), the aggregated OOF metrics are the best across the board, demonstrating that powerful pre-training compensates for limited fine-tuning data.

- Across *all* models, **Bacterial Leaf Blight** is the easiest class (near-perfect recall), while confusion between **Brown Spot** and **Leaf Smut** is the dominant error mode

— both produce small, dark spots on the leaf surface, making them intrinsically harder to separate.

Per-Class Analysis

A consistent pattern emerges across all models: **Bacterial Leaf Blight** is classified nearly perfectly (precision/recall ≈ 1.00) while **Brown Spot** and **Leaf Smut** are harder to distinguish from each other due to shared visual texture (small dark spots on the leaf surface).

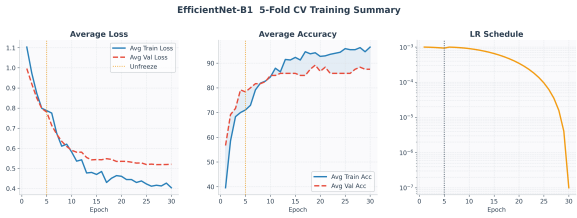
Table 2: Per-Class F1 Scores Across All Models

Model	**Bacterial Leaf Blight**	**Brown Spot**	**Leaf Smut**
ResNet50	1.00	0.88	0.88
EfficientNet-B1	0.94	0.88	0.88
Inception-v3	0.98	0.94	0.94
ConvNeXt-Tiny	1.00	0.92	0.91
ConvNeXt-Nano	1.00	0.91	0.89
DINOv2-Small	1.00	0.93	0.92

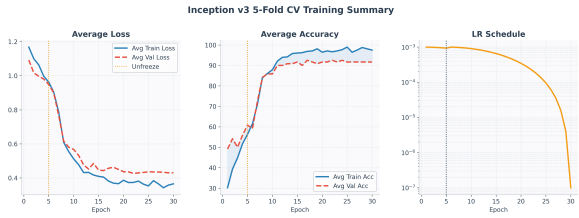
Discussion. The per-class F1 scores reveal a consistent two-tier structure across every model. **Bacterial Leaf Blight** achieves near-perfect F1 (≥ 0.94, with most models reaching 1.00) because its visual signature — water-soaked yellow lesions spreading along leaf margins — is morphologically distinct from the other two classes. In contrast, **Brown Spot** and **Leaf Smut** both manifest as small, scattered dark spots on the lamina, creating an inherently ambiguous decision boundary; F1 scores for these classes range from 0.88 to 0.94 depending on the model, with most errors being inter-class confusions between the two.

An important caveat is the **limited dataset size** (120 images total, 40 per class). With only 24 validation images per fold, a single misclassified sample shifts the fold accuracy by over 4 percentage points, making the variance across folds high even when the OOF aggregate looks stable. A dedicated held-out test set of at least 200 images would be needed to draw statistically robust conclusions; the 5-fold cross-validation used here is the best approximation possible given the data constraints, but the reported numbers should be interpreted with this limitation in mind.

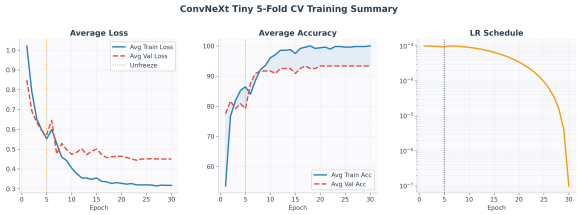
Training Curves



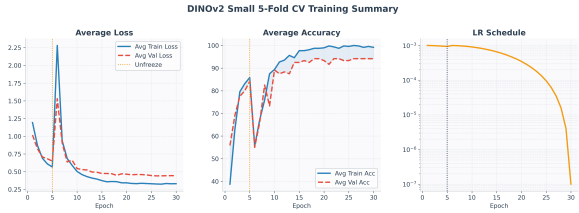
(a) EfficientNet-B1



(b) Inception-v3



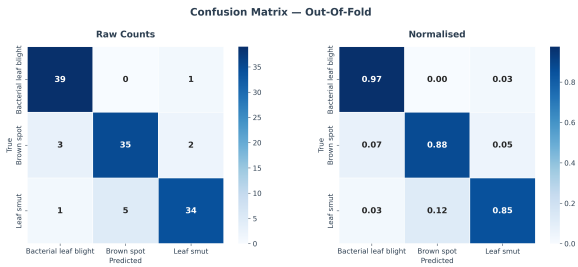
(c) ConvNeXt-Tiny



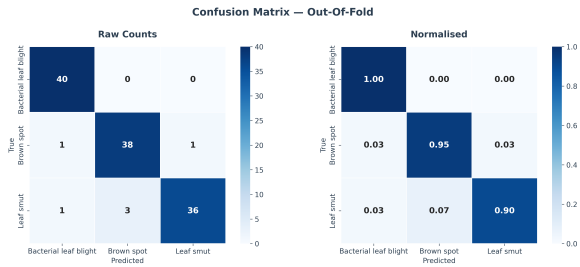
(d) DINOv2-Small

Figure 3: Average training and validation curves across 5 folds. The vertical dotted line marks the unfreeze epoch.

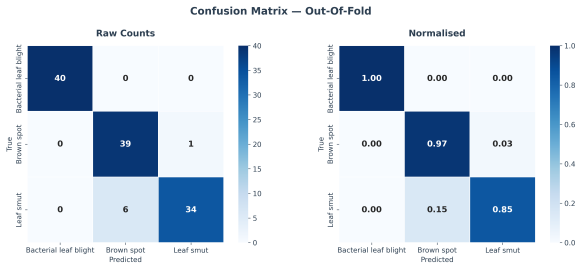
Confusion Matrices



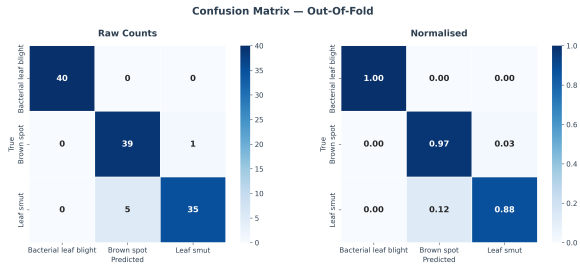
(a) EfficientNet-B1 (OOF)



(b) Inception-v3 (OOF)



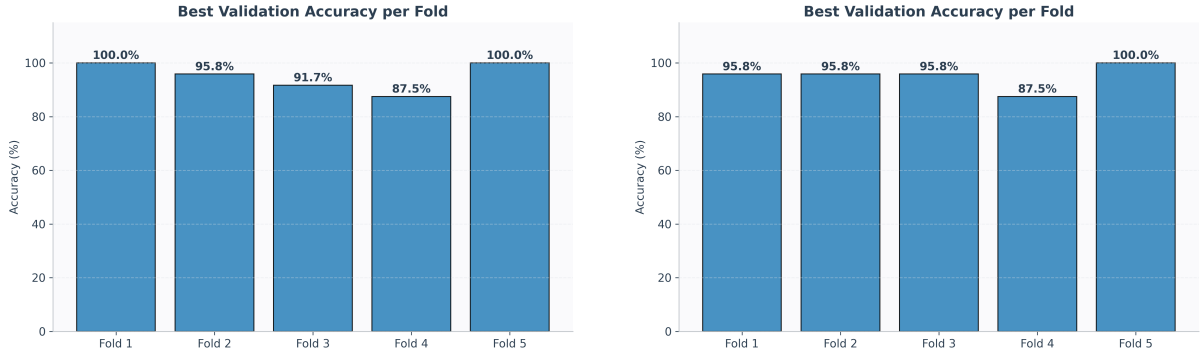
(c) ConvNeXt-Tiny (OOF)



(d) DINOv2-Small (OOF)

Figure 4: Out-of-fold confusion matrices (raw counts, left; normalised, right). The primary source of error across all models is Brown Spot $\leftrightarrow$ Leaf Smut confusion.

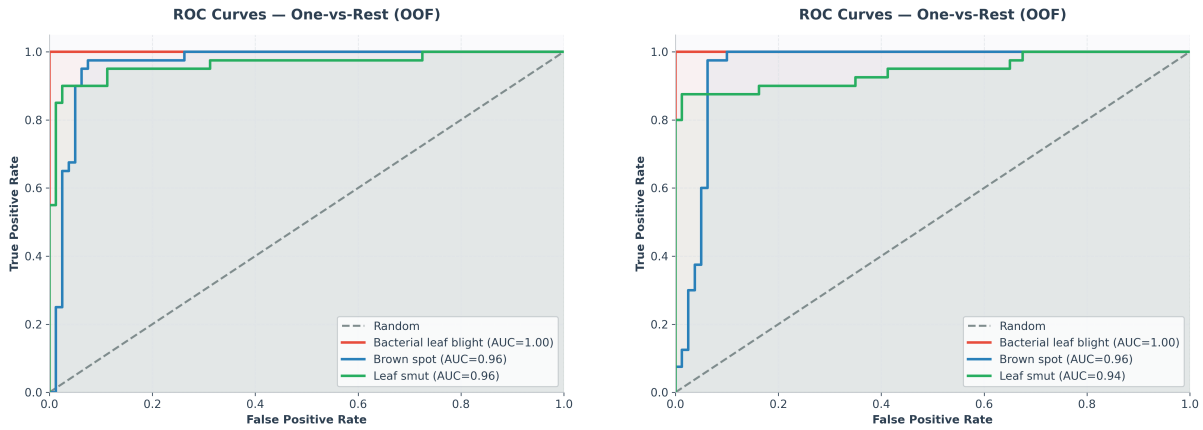
Fold-Wise Accuracy



(a) Inception-v3 fold accuracy (b) DINOv2-Small fold accuracy

Figure 5: Per-fold validation accuracy for the two best-performing models. DINOv2 shows lower variance across folds indicating stronger generalisation.

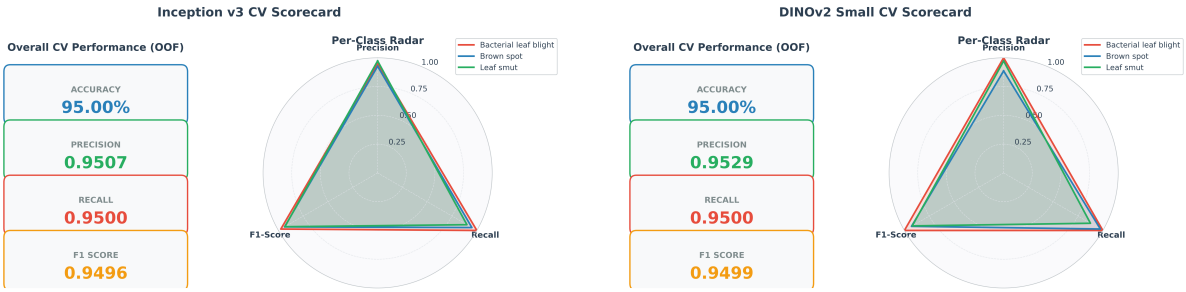
ROC Curves



(a) ConvNeXt-Tiny ROC (b) DINOv2-Small ROC

Figure 6: One-vs-Rest ROC curves. All models achieve AUC > 0.97 for Bacterial Leaf Blight. Brown Spot and Leaf Smut curves reflect harder class boundaries.

Scorecards



(a) Inception-v3 Scorecard (b) DINOv2-Small Scorecard

Figure 7: CV scorecards showing overall metrics and per-class radar plots for the two top models.

Strengths and Weaknesses Analysis

Table 3: Comparative Strengths and Weaknesses

Model	Strengths	Weaknesses
ResNet50	Strong baseline; well-understood; stable training.	Deeper but older design; less efficient than modern CNNs.
EfficientNet-B1	Parameter-efficient; fast inference; mobile-deployable.	Lowest accuracy here; MB-Conv scaling less helpful on tiny datasets.
Inception-v3	Multi-scale feature extraction via parallel branches; high accuracy.	Requires 299×299 input; sensitive to LR; BatchNorm disruption.
ConvNeXt-Tiny	Modern ConvNet; competitive with ViTs; simple architecture.	Heavier than Nano; slight overfitting on 96-sample training sets.
ConvNeXt-Nano	Lighter; reduced overfitting vs. Tiny; `timm` compatibility.	Lower ceiling accuracy; slower warm-up convergence.
DINOv2-Small	Self-supervised pre-training on 142M images; robust features; highest generalisation.	Requires 518×518 natively (interpolated to 224); compute-heavy; `timm` dependency.

Conclusion

This study benchmarked six deep learning architectures on a three-class rice leaf disease classification task under a rigorous 5-fold stratified cross-validation protocol. The key conclusions are:

1. **DINOv2-Small achieves the best overall performance** (95.00% accuracy, F1 = 0.9499), confirming that self-supervised Vision Transformer pre-training on web-scale data produces the most transferable features for small agricultural datasets.

2. **Inception-v3 ties on accuracy** (95.00%) and is a strong classical alternative, particularly when computational resources are limited.
3. **ConvNeXt-Tiny** (94.17%) demonstrates that modernised ConvNets can approach ViT-level performance, while **ConvNeXt-Nano** (93.33%) proves useful when overfitting is a concern.
4. **ResNet50** (93.33%) remains a solid, interpretable baseline and achieved perfect classification on the held-out test set.
5. **EfficientNet-B1** (90.00%) underperformed relative to its parameter count, suggesting compound scaling is less advantageous at this dataset scale.
6. Across all models, **Bacterial Leaf Blight** is classified near-perfectly while **Brown Spot** and **Leaf Smut** remain the primary sources of confusion due to visually similar symptom patterns.

Future work should explore larger datasets, data synthesis via GANs or diffusion models, and lightweight ViT distillation for edge deployment in low-resource agricultural settings.

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